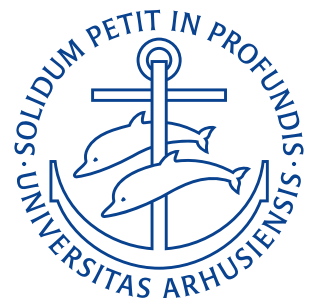


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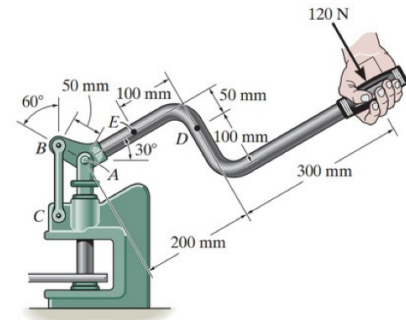
Take-Home Assignment W. 39–40

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Course: Statics & Strength of Materials — 290221U006
Date: October 22, 2025



Problem P10-1

Determine the resultant internal loadings acting on the cross section at point E of the handle arm, and on the cross section of the short link BC .



Derivation

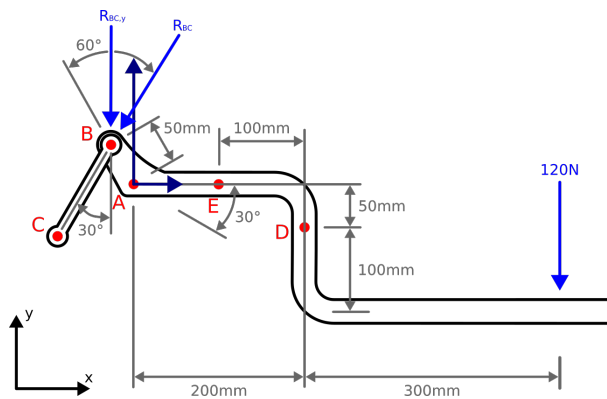


Figure 0.1: Free body diagram of the hole punch.

We start by determining the force F_{BC} in the link BC using a moment equilibrium about A , using Figure 0.1 we get

$$R_{BC} \cdot \cos 30^\circ \cdot 50 \text{ mm} - 120 \text{ N} \cdot 500 \text{ mm} = 0$$

$$R_{BC} = \frac{120 \text{ N} \cdot 500 \cdot 10^{-3} \text{ m}}{\cos 30^\circ \cdot 50 \cdot 10^{-3} \text{ m}}$$

$$= 1,39 \text{ kN}.$$

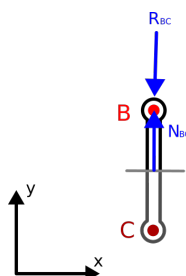


Figure 0.2: Free body diagram of BC

From Figure 0.2 we see that

$$-R_{BC} + N_{BC} = 0 \implies R_{BC} = N_{BC} = 1,39 \text{ kN}.$$

This link will not experience any shear force, torque or bending moment and the internal loadings are therefore all found.

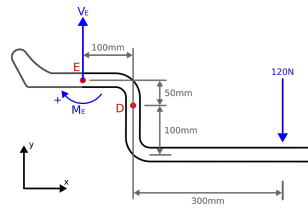


Figure 0.3: Free body diagram of the segment

From Figure 0.3 we see that

$$\sum F_x = 0 \Rightarrow N_E = 0.$$

Likewise, in the y -direction we get

$$\sum F_y = 0 \Rightarrow V_E = 120 \text{ N}.$$

The moment in E M_E must balance the moment from the applied force

$$-M_E - 120 \text{ N} \cdot 400 \text{ mm} = 0 \Rightarrow M_E = 120 \text{ N} \cdot 400 \cdot 10^{-3} \text{ m} = 48 \text{ N m}.$$

$$N_{BC} = 1,39 \text{ kN}$$

$$V_E = 120 \text{ N}$$

$$M_E = 48 \text{ N m}$$

RESULT

Problem P10-2

The bar has a cross-sectional area of $400 \cdot 10^{-6} \text{ m}^2$. If it is subjected to a triangular axial distributed loading along its length which is 0 at $x = 0$ and $9 \frac{\text{kN}}{\text{m}}$ at $x = 1,5 \text{ m}$, and to two concentrated loads as shown, determine the average normal stress in the bar as a function of x for $0 \leq x \leq 0,6 \text{ m}$.

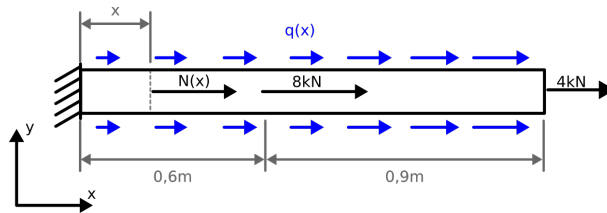
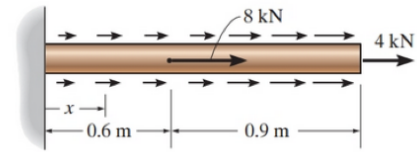


Figure 0.4: Free body diagram of the bar.

Derivation

We consider the right segment of the bar from x to $1,5 \text{ m}$. A force balance in the x -direction with positive to the right yields

$$\begin{aligned} N &= 8 \text{ kN} + 4 \text{ kN} + \frac{1}{2} \left(6 \frac{\text{kN}}{\text{m}} \cdot x + 9 \text{ kN} \right) (1,5 \text{ m} - x) \\ &= 18,8 \text{ kN} - 3 \frac{\text{kN}}{\text{m}^2} \cdot x^2. \end{aligned}$$

The average normal stress is simply defined as

$$\sigma_{\text{avg}} = \frac{N}{A},$$

hence,

$$\begin{aligned} \sigma_{\text{avg}}(x) &= \frac{18,8 \text{ kN}}{400 \cdot 10^{-6} \text{ m}^2} - \frac{3 \frac{\text{kN}}{\text{m}^2}}{400 \cdot 10^{-6} \text{ m}^2} x^2 \\ &= 46,9 \text{ MPa} - 7,5 \frac{\text{MPa}}{\text{m}^2} \cdot x^2. \end{aligned}$$

$$\sigma_{\text{avg}}(x) = 46,9 \text{ MPa} - 7,5 \frac{\text{MPa}}{\text{m}^2} \cdot x^2$$

RESULT

Problem P11-1

Determine the required thickness of member BC to the nearest mm, and the diameter of the pins at A and B if the allowable normal stress for member BC is $\sigma_{\text{allow}} = 200 \text{ MPa}$ and the allowable shear stress for the pins is $\tau_{\text{allow}} = 70 \text{ MPa}$.

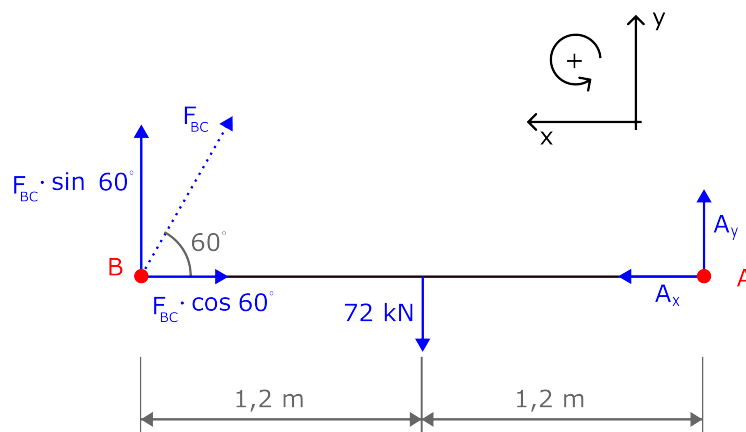
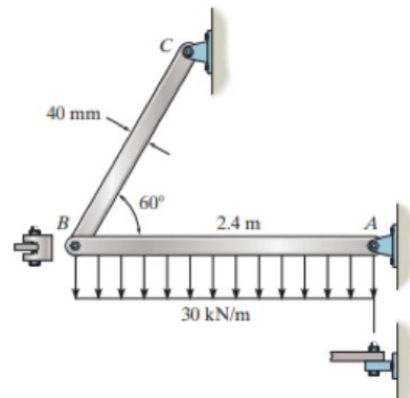


Figure 0.5: Free body diagram of the structure.

Derivation

We start by converting the distributed load into an equivalent point load, this load will be

$$P = 30 \frac{\text{kN}}{\text{m}} \cdot 2,4 \text{ m} = 72 \text{ kN}.$$

This load will be placed at the middle of the member, 1,2 m from A as shown on Figure 0.5. We start with the moment equilibrium about point A as

$$\begin{aligned} \sum M_A &= 0 \\ 0 &= P \cdot 1,2 \text{ m} - F_{BC} \cdot \sin 60^\circ \cdot 2,4 \text{ m} \\ F_{BC} &= \frac{72 \text{ kN} \cdot 1,2 \text{ m}}{\sin 60^\circ \cdot 2,4 \text{ m}} \\ &= 41,6 \text{ kN}. \end{aligned}$$

Now a force equilibrium in the x -direction yields

$$\begin{aligned} \sum F_x &= 0 \\ 0 &= F_{BC} \cdot \cos 60^\circ - A_x \\ A_x &= F_{BC} \cdot \cos 60^\circ \end{aligned}$$

$$= 41,6 \text{ kN} \cdot \cos 60^\circ = 20,8 \text{ kN}.$$

In the y -direction we get

$$\begin{aligned}\sum F_y &= 0 \\ 0 &= F_{BC} \cdot \sin 60^\circ + A_y - P \\ A_y &= F_{BC} \cdot \sin 60^\circ - P \\ &= 72 \text{ kN} - 41,6 \text{ kN} \cdot \sin 60^\circ = 36 \text{ kN}.\end{aligned}$$

This corresponds to a total reaction at A of

$$R_A = \sqrt{A_x^2 + A_y^2} = \sqrt{(20,8 \text{ kN})^2 + (36 \text{ kN})^2} = 41,6 \text{ kN}.$$

The required thickness of BC can be found from the allowable normal stress as

$$\begin{aligned}\sigma_{\text{allow}} &= \frac{F_{BC}}{A_{BC}} \\ A_{BC} &= \frac{F_{BC}}{\sigma_{\text{allow}}} \\ &= \frac{41,6 \text{ kN}}{200 \text{ MPa}} \\ &= 2,08 \cdot 10^{-4} \text{ m}^2 \\ A_{BC} &= w_{BC} \cdot h_{BC} \\ w_{BC} &= \frac{A_{BC}}{h_C} \\ &= \frac{2,08 \cdot 10^{-4} \text{ m}^2}{0,04 \text{ m}} \\ &= 5,2 \text{ mm}.\end{aligned}$$

A can be seen to be a single-stress shear pin on the figure, hence

$$\begin{aligned}\tau_{\text{allow}} &= \frac{R_A}{A_A} \\ A_A &= \frac{R_A}{\tau_{\text{allow}}} \\ &= \frac{41,6 \text{ kN}}{70 \text{ MPa}} \\ &= 5,9 \cdot 10^{-4} \text{ m}^2 \\ A_A &= \pi \cdot \frac{D_A^2}{4} \\ D_A &= \sqrt{\frac{4 \cdot A_A}{\pi}} \\ &= \sqrt{\frac{4 \cdot 5,9 \cdot 10^{-4} \text{ m}^2}{\pi}} \\ &= 27 \text{ mm}.\end{aligned}$$

And as B is a double-stress shear pin we get

$$\begin{aligned}\tau_{\text{allow}} &= \frac{F_{BC}}{2A_B} \\ A_B &= \frac{2 \cdot \tau_{\text{allow}}}{F_{BC}} \\ &= \frac{41,6 \text{ kN}}{2 \cdot 70 \text{ MPa}}\end{aligned}$$

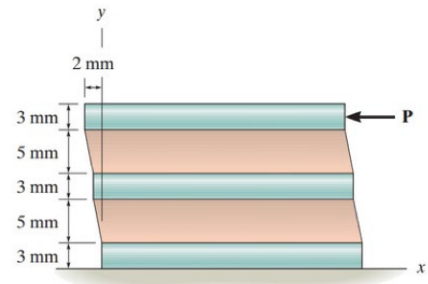
$$\begin{aligned} &= 3 \cdot 10^{-4} \text{ m}^2 \\ A_B &= \pi \cdot \frac{D_B^2}{4} \\ D_B &= \sqrt{\frac{4 \cdot A_B}{\pi}} \\ &= \sqrt{\frac{4 \cdot 3 \cdot 10^{-4} \text{ m}^2}{\pi}} \\ &= 19 \text{ mm.} \end{aligned}$$

$w_{BC} = 5,2 \text{ mm}$
$D_A = 27 \text{ mm}$
$D_B = 19 \text{ mm.}$

RESULT

Problem P11-2

Nylon strips are fused to glass plates. When moderately heated the nylon will become soft while the glass stays approximately rigid. Determine the average shear strain in the nylon due to the load P when the assembly deforms as indicated.



Derivation

As the glass is approximately rigid we can assume it does not deform at all and that all of the material deformation is due to the nylon deforming.

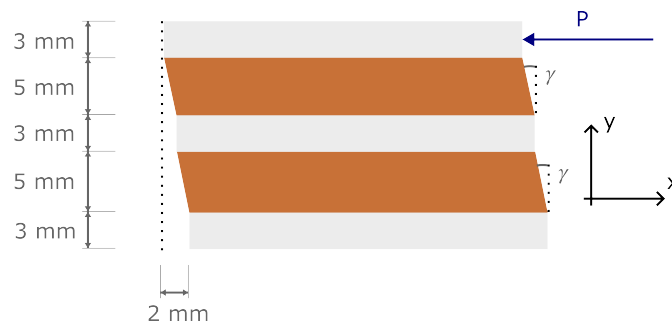


Figure 0.6: Free body diagram of the system.

Shear strain is defined as the angle the material deviates from perpendicular. This can be thought of as a right triangle giving:

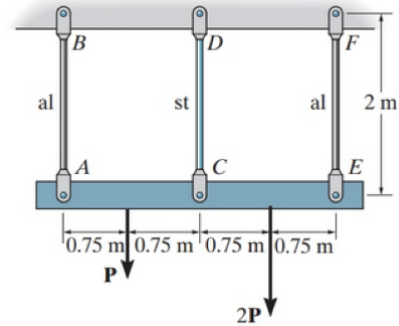
$$\begin{aligned}\tan \gamma &= \frac{2 \text{ mm}}{10 \text{ mm}} \\ \gamma &= \tan^{-1} \left(\frac{2}{10} \right) \\ &= 0,197.\end{aligned}$$

$$\boxed{\gamma = 0,197}$$

RESULT

Problem P12-1

The rigid beam is supported by the three suspender bars. Bars AB and EF are made of aluminum and bar CD is made of steel. If each bar has a cross-sectional area of 450 mm^2 , determine the maximum value of P if the allowable stress is $(\sigma_{\text{allow}})_{\text{st}} = 200 \text{ MPa}$ for the steel and $(\sigma_{\text{allow}})_{\text{al}} = 150 \text{ MPa}$ for the aluminum. $E_{\text{st}} = 200 \text{ GPa}$, $E_{\text{al}} = 70 \text{ GPa}$.



Derivation

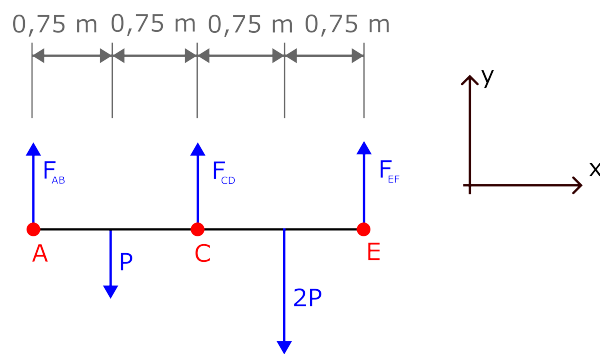


Figure 0.7: Free body diagram of the beam.

We start by considering the moment about A giving us

$$\begin{aligned}\sum M_A &= 0 \\ 0 &= -P \cdot 0.75 \text{ m} + F_{CD} \cdot 1.5 \text{ m} - 2P \cdot 2.25 \text{ m} + F_{EF} \cdot 3 \text{ m} \\ 5.25 \text{ m} \cdot P &= 1.5 \text{ m} \cdot F_{CD} + 3.0 \text{ m} \cdot F_{EF}.\end{aligned}$$

Using the moment equilibrium about E yields,

$$\begin{aligned}\sum M_E &= 0 \\ 0 &= 2P \cdot 0.75 \text{ m} - F_{CD} \cdot 1.5 \text{ m} + P \cdot 2.25 \text{ m} - F_{AB} \cdot 3 \text{ m} \\ 3.75 \text{ m} \cdot P &= 1.5 \text{ m} \cdot F_{CD} + 3.0 \text{ m} \cdot F_{AB}.\end{aligned}$$

Using the fact that the bar is rigid the total system must deform as shown on [Figure 0.8](#).

From similar triangles we have that:

$$\begin{aligned}\frac{\delta_{CD} - \delta_{AB}}{1.5} &= \frac{\delta_{EF} - \delta_{AB}}{3} \\ 3\delta_{CD} - 3\delta_{AB} &= 1.5\delta_{EF} - 1.5\delta_{AB} \\ 3\delta_{CD} &= 1.5\delta_{EF} + 1.5\delta_{AB}\end{aligned}$$

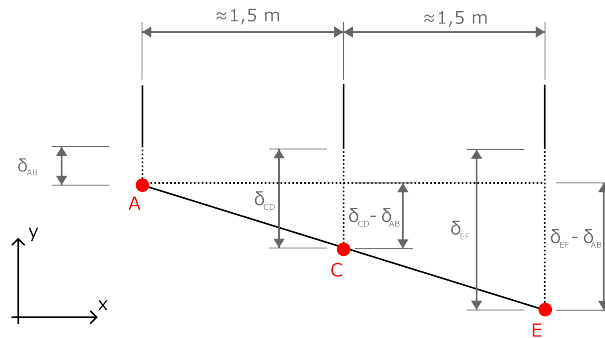


Figure 0.8: Deformation of the system

$$2\delta_{CD} = \delta_{EF} + \delta_{AB}.$$

Now we can use the formula for deformation

$$\delta = \frac{PL}{AE}$$

and get

$$\begin{aligned} \frac{2F_{CD} \cdot L}{A \cdot E_{st}} &= \frac{L}{A \cdot E_{al}} (F_{EF} + F_{AB}) \\ F_{CD} &= \frac{E_{st}}{2 \cdot E_{al}} (F_{EF} + F_{AB}) \\ &= \frac{200 \text{ GPa}}{2 \cdot 70 \text{ GPa}} (F_{EF} + F_{AB}) \\ &= \frac{10}{7} (F_{EF} + F_{AB}). \end{aligned}$$

We now have three equations with three unknowns:

$$\begin{aligned} 1,5 \cdot F_{CD} + 3 \cdot F_{EF} &= 5,25P \\ 1,5 \cdot F_{CD} + 3 \cdot F_{AB} &= 3,75P \\ F_{CD} &= \frac{10}{7} (F_{EF} + F_{AB}). \end{aligned}$$

Solving these gives:

$$\begin{aligned} F_{AB} &= 0,368P \\ F_{CD} &= 1,76P \\ F_{EF} &= 0,868P. \end{aligned}$$

Now the maximally allowable value of P for each beam can be found separately using the following method

$$\begin{aligned} (\sigma_{\text{allow}})_{\text{al}} &= \frac{F_{AB}}{A} \\ F_{AB} &= (\sigma_{\text{allow}})_{\text{al}} \cdot A \\ 0,368 (P_{\text{allow}})_{AB} &= 150 \text{ MPa} \cdot 450 \cdot 10^{-6} \text{ m}^2 \\ (P_{\text{allow}})_{AB} &= \frac{150 \text{ MPa} \cdot 450 \cdot 10^{-6} \text{ m}^2}{0,368} \\ &= 184 \text{ kN}. \end{aligned}$$

Doing the same for the other suspenders yields:

$$(P_{\text{allow}})_{CD} = 77,8 \text{ kN}$$

$$(P_{\text{allow}})_{EF} = 51,0\text{kN}.$$

And from the above results it becomes obvious that suspender EF will limit the allowable load to $P_{\text{max}} = (P_{\text{allow}})_{EF} = 51,0\text{kN}$.

$P_{\text{max}} = 51,0\text{kN}$

RESULT